

Project Proposal: The Little Chef's Library Database

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Project Objectives

The Little Chef's Library Database aims to create an integrated system that bridges children's literacy with culinary education by establishing meaningful connections between books and cooking experiences. The primary objective are to:

1. Facilitate Discovery - Enable parents and educators to seamlessly find recipes tied to specific books and identify books featuring particular foods or cuisines.
2. Track Development - Monitor children's progress and preferences across both reading comprehension and cooking skills development.
3. Streamline Operations - Manage event logistics for cooking sessions and maintain comprehensive records of program participation.
4. Support Personalization - Account for individual dietary restrictions and reading levels to provide tailored recommendations.

Application / Use Case

This database serves a critical real-world need in educational programming where literacy and life skills intersect. Libraries, schools, and community centers running combined reading-cooking programs currently face challenges in manually tracking book-recipe relationships, participant progress, and session logistics. The system provides immediate practical value by:

- Helping educators quickly plan cooking activities that reinforce literary themes
- Allowing parents to extend reading experiences into hands-on kitchen learning
- Enable program administrators to demonstrate participant growth and program effectiveness through measurable outcomes

Proposed Features / Functional Requirements

Core User Functions:

- Book-Recipe Matching: Searching for recipes based on recently read books and vice versa
- Participant Management: Track child profiles including reading levels, dietary restrictions, and cooking experience
- Session Coordination: Schedule cooking sessions, assign participants, and record outcome
- Theme-Based Discovery: Find books and recipes by educational themes or culinary categories
- Progress Monitoring: Generate reports on individual and group development in both literacy and cooking skills

System Features:

- Advanced Search: Multi-criteria filtering by age group, difficulty level, dietary needs, and reading level
- Allergen Alert System: Automatic flagging of recipes containing common allergens based on participant profiles
- Session Planning Tools: Capacity management and ingredient requirement forecasting

Important Data Description

Primary Data Sources:

- Book Metadata - ISBN, title, author, age appropriateness
- Recipe Information - Ingredients, instructions, difficulty, prep time
- Participant Profiles- Demographic data, restrictions, skill levels
- Session Records - Attendance, outcomes, feedback

Critical Data Types & Justification

- Structure Dietary Data - Essential for safety and inclusion, using standardized allergen categories
- Skill Progression Metrics - Both reading levels and cooking difficulty scales required for age appropriate matching
- Theme Taxonomy - Controlled vocabulary to ensure consistent categorization for effective discovery

Project Feasibility & Plan

- Sprint 1: Database Foundation (Week 1-3): Requirements analysis, ER Modeling, logical design, normalization, physical schema implementation
- Sprint 2: Core Functionality (Weeks 4-6): Design Validation, test data loading, critical query implementation, performance tuning, documentation

Task Division

- Design and Architecture: Conceptual modeling, logical design, physical implementation
- Development and Implementation: DDL scripting, query development, data population
- Testing and Documentation: Performance optimization, technical documentation